

Discrete Mathematics Spring 2026
Homework 2: Propositional Logic (PL)
Due Sunday February 8 @11:59:00pm Eastern

Show all your work to receive full credit

1. Truth Table

Write a truth table for the following proposition. Show all the intermediary columns to receive full credit.

$$(p \wedge \neg q) \rightarrow r$$

2. Propositions in PL and Negation

Let c , m , d , and h be the following propositions:

- c : The car is red
- m : The car is a manual transmission
- d : The car is a diesel engine
- h : The car has high mileage

(a) Express each of these propositions as an English sentence.

1. $\neg c \rightarrow \neg h$
2. $c \rightarrow (\neg m \vee \neg d)$
3. $(c \wedge m) \rightarrow \neg d$

(b) What is the negation of the first two propositions in propositional logic? Make sure to push the negation inside using De Morgan's laws and simplify.

3. Implication

Consider the following implication:

If the temperature is below freezing then the lake is frozen

1. Write in English the converse, inverse, and the contrapositive of the implication above.
2. Write the negation of the implication in English. Express the negation without using phrases like "It's not true that..." or "It is not the case that..."

4. Laws of Propositional Logic Using the laws of PL (not truth tables), prove the following statements.

1. $(p \vee q) \wedge (\neg p \wedge \neg q) \equiv F$ (in other words, the proposition is a fallacy).
2. $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$
3. $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r) \equiv T$ (in other words, the proposition is a tautology).

Note: No need to justify each step with the name of the law used. Please use a new line for each step for clarity.

5. **Rewriting the Implication** Consider the proposition: If I'm not tired and not busy then I will go to the party.

Translated in propositional logic as follows:

$$(\neg \text{tired} \wedge \neg \text{busy}) \rightarrow \text{go}$$

Rewrite the proposition without $\rightarrow$ and using only $\vee$, $\wedge$, and $\neg$ as needed. Simplify your answer so that no parentheses are required.

I have NO idea what I'm doing...



but at least my truth table looks right.

Credit: Image LLM-generated